

An Affirmative Answer to Question 10.6 on Linear Kalton Maps

Candidate full solution packet for run `fa_banach_001`

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Status. Candidate full solution, verdict *likely valid*, pending independent human review.

1 Source problem

Dikranjan, Shakhmatov, and Spěvák define the integer Kalton map

$$K_A : \bigoplus_{a \in A} \langle a \rangle \longrightarrow E$$

and its real-linear extension

$$\ell K_A : \bigoplus_{a \in A} \mathbb{R}a \longrightarrow E.$$

In Section 10, Question 10.6, they ask whether a topologically isomorphic embedding K_A forces ℓK_A to be a topologically isomorphic embedding as well [1].

If A is a subset of a topological vector space such that the Kalton map $\mathcal{K}_A$ is topologically isomorphic embedding, then the set A is topologically independent (Proposition 4.7) and consequently, it is also linearly independent by Proposition 4.11(i). Thus the linear Kalton map $\ell \mathcal{K}_A$ is an isomorphism onto $\langle A \rangle_{\mathbb{R}}$. Moreover, $\ell \mathcal{K}_A$ is also continuous by Proposition 10.1. However, we do not know whether it is an open map onto its image $\langle A \rangle_{\mathbb{R}}$.

Question 10.6. If A is a subset of a topological vector space such that $\mathcal{K}_A$ is a topologically isomorphic embedding, is then $\ell \mathcal{K}_A$ a topologically isomorphic embedding as well?

Figure 1: Question 10.6 and its immediate context in the source paper, PDF page 20.

Equivalently, their preceding results already give that ℓK_A is an injective continuous linear map, and the remaining issue is whether it is open onto $\text{span}_{\mathbb{R}} A$.

2 Definitions and source reductions

All topological vector spaces below are Hausdorff and real. For a complex space, one may pass to the underlying real space.

A set $A \subset E \setminus \{0\}$ is *topologically independent* if for every neighbourhood W of 0 there is a neighbourhood V of 0 such that, for every finitely supported family $(z_a)_{a \in A}$ of integers,

$$\sum_{a \in A} z_a a \in V \implies z_a a \in W \quad (a \in A).$$

It is *absolutely Cauchy summable* if every neighbourhood U of 0 contains the subgroup generated by $A \setminus F$ for some finite $F \subset A$.

We use four results proved in the source [1]:

1. K_A is a topologically isomorphic embedding if and only if A is topologically independent and absolutely Cauchy summable (Theorem 5.1).
2. A topologically independent subset of a real topological vector space is linearly independent (Proposition 4.11(i)).
3. If A is absolutely Cauchy summable, then for every neighbourhood U of 0 there is a finite $F \subset A$ such that

$$\text{span}_{\mathbb{R}}(A \setminus F) \subset U \tag{1}$$

(Proposition 9.2(i)), and ℓK_A is continuous (Proposition 10.1).

4. For linearly independent A , the linear Kalton map is open onto its image if and only if every canonical coordinate projection

$$\pi_a^A : \text{span}_{\mathbb{R}} A \longrightarrow \mathbb{R}a$$

is continuous (Proposition 10.2).

Thus it suffices to prove continuity of every π_a^A .

3 Proof intuition

The obstruction exhibited by a discontinuous real coordinate is a sequence of very small real combinations whose distinguished coefficient stays equal to one. Topological independence only tests integer combinations, so the key is to round a suitable multiple of such a real combination.

The rounding is split into two regions. Absolute Cauchy summability puts the entire real span of a cofinite tail inside a prescribed small neighbourhood, so all tail rounding errors are harmless regardless of their size. Only a finite head remains. Simultaneous Dirichlet approximation rounds those finitely many coefficients with a uniformly bounded denominator. The allegedly discontinuous vector is chosen only after that denominator bound is known, so every possible multiple remains small. The rounded integer combination then contradicts topological independence at the distinguished coordinate.

4 A simultaneous rounding lemma

Lemma 1. *Let $r_1, \dots, r_m \in \mathbb{R}$ and let $Q \geq 2$ be an integer. There exist an integer q with $1 \leq q \leq Q^m$ and integers $z_1, \dots, z_m$ such that*

$$|qr_j - z_j| \leq \frac{1}{Q} \quad (1 \leq j \leq m).$$

For $m = 0$, take $q = 1$.

Proof. For $m \geq 1$, place the $Q^m + 1$ points

$$(\{kr_1\}, \dots, \{kr_m\}), \quad 0 \leq k \leq Q^m,$$

in the Q^m half-open boxes of side length $1/Q$ partitioning $[0, 1)^m$. Two points lie in the same box. Subtract their indices and the corresponding integer parts. The positive index difference is at most Q^m and has the asserted property. $\square$

5 Main theorem and proof

Theorem 1. *Let E be a Hausdorff real topological vector space and let $A \subset E \setminus \{0\}$. If the integer Kalton map K_A is a topologically isomorphic embedding, then the linear Kalton map ℓK_A is a topologically isomorphic embedding.*

Proof. By the source reductions, A is topologically independent, absolutely Cauchy summable, and linearly independent; moreover, ℓK_A is an injective continuous linear map. It remains to prove that it is open onto $\text{span}_{\mathbb{R}} A$.

Fix $a_0 \in A$. Write

$$\pi_{a_0}^A(x) = \varphi(x)a_0 \quad (x \in \text{span}_{\mathbb{R}} A),$$

where φ is the coefficient functional at a_0 . Suppose for a contradiction that φ is discontinuous.

The cyclic subgroup $\mathbb{Z}a_0$ is discrete in the one-dimensional subspace $\mathbb{R}a_0$. Hence there is a neighbourhood W of 0 in E such that

$$W \cap \mathbb{Z}a_0 = \{0\}. \quad (2)$$

Let V be a W -witness for the topological independence of A . Choose neighbourhoods U_1, U_2, U_3 of 0, with U_1 balanced, such that

$$U_1 + U_2 + U_3 \subset V. \quad (3)$$

Apply (1) to U_3 . After adjoining a_0 to the exceptional set, we obtain a finite $F \subset A$ with $a_0 \in F$ and

$$\text{span}_{\mathbb{R}}(A \setminus F) \subset U_3. \quad (4)$$

Put $H = F \setminus \{a_0\}$ and $m = |H|$. The finite-dimensional map

$$\mathbb{R}^H \longrightarrow E, \quad (t_b)_{b \in H} \longmapsto \sum_{b \in H} t_b b,$$

is continuous. Therefore there is $\eta > 0$ such that

$$|t_b| < \eta \ (b \in H) \implies \sum_{b \in H} t_b b \in U_2. \quad (5)$$

Choose an integer $Q \geq 2$ with $1/Q < \eta$, and set $N = Q^m$ (with $N = 1$ when $m = 0$).

A discontinuous nonzero linear functional takes every real value on every neighbourhood of zero: otherwise its image on some neighbourhood would be a proper balanced subset of $\mathbb{R}$ and hence bounded, forcing continuity. Apply this to the neighbourhood $(N^{-1}U_1) \cap \text{span}_{\mathbb{R}} A$. There is a finitely supported real combination

$$x = \sum_{b \in A} r_b b \in N^{-1}U_1, \quad r_{a_0} = \varphi(x) = 1. \quad (6)$$

Apply Lemma 1 to $(r_b)_{b \in H}$. We obtain $1 \leq q \leq N$ and integers z_b for $b \in H$ satisfying

$$|z_b - qr_b| < \eta \quad (b \in H). \quad (7)$$

Set $z_{a_0} = q$. For each $b \in A \setminus F$ in the finite support of x , choose an integer z_b nearest to qr_b ; set all other z_b equal to zero. The integer combination $y = \sum_b z_b b$ has finite support, and

$$y = qx + \sum_{b \in H} (z_b - qr_b)b + \sum_{b \in A \setminus F} (z_b - qr_b)b.$$

Because $x \in N^{-1}U_1$, $q \leq N$, and U_1 is balanced, the first term belongs to U_1 . The second belongs to U_2 by (5)–(7). The third is an element of $\text{span}_{\mathbb{R}}(A \setminus F)$ and hence belongs to U_3 by (4). Thus (3) gives $y \in V$.

Since V is a W -witness, topological independence forces $z_{a_0}a_0 = qa_0 \in W$. This contradicts (2), because $q \geq 1$. Therefore φ , and hence $\pi_{a_0}^A$, is continuous.

The choice of a_0 was arbitrary, so every canonical coordinate projection is continuous. Proposition 10.2 of the source now makes ℓK_A open onto its image. Together with its injectivity and continuity, this proves that ℓK_A is a topologically isomorphic embedding. $\square$

6 Verification notes

1. The order of choices is essential. The tail neighbourhood and finite set F are fixed first; the denominator bound N is then determined by the finite head; only afterward is x chosen in $N^{-1}U_1$. Thus no circular dependence is hidden in the proof.
2. The tail estimate controls the full real span, so the tail rounding errors need no coefficient bound. This is the precise role of absolute Cauchy summability beyond continuity of ℓK_A .
3. All combinations are finitely supported. No convergence or completeness of E is used.
4. The distinguished coefficient is rounded without error because $r_{a_0} = 1$ and $z_{a_0} = q$.
5. No computational verification is relevant. The separate `verification_report.md` records an adversarial step check.

7 Novelty check and limitations

The local run indexes were searched for arXiv:1508.00667, the exact wording of Question 10.6, *linear Kalton map*, *topologically independent*, and *absolutely Cauchy summable*. A bounded external search used the exact question, paper title and authors, and close variants involving integer and linear Kalton embeddings. It returned the source records but no later paper claiming to answer Question 10.6. The search was not an exhaustive citation audit, so novelty confidence is moderate.

The theorem answers Question 10.6. It does not settle Question 9.8 of the source for arbitrary absolutely Cauchy summable sets lacking topological independence.

8 Human-review recommendation

The result should be sent to human review as a candidate full solution. The main verification target is the decomposition of the rounded integer combination into the bounded multiple of x , the finite-head approximation error, and the arbitrary-coefficient tail error.

References

- [1] D. Dikranjan, D. Shakhmatov, and J. Spěvák, *Direct sums and products in topological groups and vector spaces*, J. Math. Anal. Appl. **437** (2016), 1257–1282; [arXiv:1508.00667](https://arxiv.org/abs/1508.00667).